Karan Kumar

Roll No.: 230102036
B. Tech
Electronics and Communication Engineering
Indian Institute of Information Technology, Bhagalpur

+91-7250064637
Karan.230102036@iiitbh.ac.in
krnroyy@gmail.com
GitHub: Krnroyy
LinkedIn: krnroyy

Professional Summary

ECE undergraduate graduating in 2027 with experience in **RTL design, FPGA/SoC integration, digital verification, and CMOS simulation**. Built AES-GCM systems on ZCU104, SystemVerilog/UVM testbenches, formal FIFO checks, and transistor-level circuits using Cadence Virtuoso.

Education

- **Indian Institute of Information Technology, Bhagalpur** — 2023 – 2027
  B. Tech in Electronics and Communication Engineering — CGPA: 7.95
- **U.R College Rosera, Samastipur, Bihar, 12th** — 2020 – 2022
  Bihar School Examination Board — Percentage: 86.0
- **Simultala Awasiya Vidyalaya, Simultala, Bihar, 10th** — 2020
  Bihar School Examination Board — Percentage: 92.8

Technical Skills

- **RTL and Verification:** Verilog, SystemVerilog, UVM Fundamentals, Testbenches, Scoreboards, Assertions, Directed and Randomized Testing
- **FPGA and SoC:** Vivado, Vitis, XSIM, ZCU104, Zynq UltraScale+ MPSoC, Bare-Metal C, AXI4-Lite, AXI4-Stream, AXI DMA
- **Simulation and Formal:** Icarus Verilog, GTKWave, Yosys/SMTBMC, Z3, Synthesis, Implementation, Timing and Resource Analysis
- **CMOS Design:** Cadence Virtuoso, Spectre, gpdk180, Hierarchical Schematics, Transient Simulation, Delay and Power Measurement
- **Programming and Workflow:** C, C++, Python, Git, GitHub Actions, Linux/WSL; UART, BRAM, Memory-Mapped Registers

Technical Projects

- **AES-128-GCM Image Encryption Accelerator on ZCU104** — *2026*
  Verilog, Vivado, Vitis, AXI DMA, Cortex-A53, Python — [GitHub]
  - Integrated AES-GCM RTL with AXI DMA and Cortex-A53 software; achieved **11/11 ciphertext and tag matches** against Python AESGCM, exact image recovery, and modified-ciphertext rejection in documented board tests.
  - Optimized protected buffer storage from **58,392 LUTs to 86 LUTs and 2 RAMB36 blocks**; the optimized build met **75.002 MHz** timing with **+0.975 ns WNS**.
  - Implemented authentication before plaintext release using a **4 KB on-chip buffer**; validated recovery and ciphertext/AAD tamper rejection for **17×19, 256×256, and 512×512** images on ZCU104.
- **AXI4-Stream FIFO: Simulation and Formal Verification** — *2026*
  SystemVerilog, Icarus Verilog, Yosys/SMTBMC, Z3 — [GitHub]
  - Designed a parameterized synchronous FIFO preserving **TDATA, TKEEP, and TLAST**; achieved **12/12 passing simulation runs** across depths **1, 3, 4, and 16**, with **32,153 output beats** checked by a scoreboard.
  - Verified **15 safety assertions under 3 assumptions** using base checks and induction for an **8-bit, depth-4** configuration; reached **3 cover scenarios**, detected an injected handshake bug, and automated checks through GitHub Actions.
- **AXI4-Lite Timer with UVM Verification** — *2026*
  SystemVerilog, UVM, Vivado Simulator, AXI4-Lite — [GitHub]
  - Implemented a countdown timer with **4 memory-mapped registers**, byte-write strobes, busy status, and interrupts; verified reload, countdown completion, and interrupt clearing.
  - Built a UVM environment with sequence, sequencer, driver, monitor, and scoreboard; verified reset during an incomplete write and repeated timer runs, with **zero reported UVM warnings, errors, or fatal errors**.
- **CMOS Full Adder Design and Load Characterization** — *Oct 2026*
  Cadence Virtuoso, Spectre, gpdk180 — [GitHub]
  - Designed a hierarchical **44-transistor CMOS full adder** using NAND/XOR cells in **180 nm CMOS**; verified all **8 input combinations** through schematic-level simulation at **1.8 V**.
  - Characterized **5–50 fF loads per output**; measured selected Cin-to-Sum rising/falling delays of **114.6/159.7 ps** and **20.8 μW** average supply power at **10 fF per output**, NN corner, 27°C.

Additional Experience

- **Video Editor Intern – PhysicsWallah** — Jan 2025 – Present
  Produced educational videos for a **50K+ subscriber** YouTube channel, meeting production deadlines.
- **Sales Associate – Unacademy** — Apr 2021 – Sep 2022
  Generated **INR 20L+** in sales through multi-channel outreach and customer communication.

Achievements & Leadership

- **Finalist, Ideastorm E-Summit 2024, IIT Roorkee**, for presenting a business solution.
- **Board Games Secretary, IIIT Bhagalpur** (Aug 2025 – 2026); completed the **32-hour Art of Communication** program.